DEEPAK KATUKURI

AI AUTOMATION ENGINEER · LLM SYSTEMS · APPLIED AI



CONTACT

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TECHNICAL SKILLS

LLM & Agentic AI:
Retrieval-Augmented Generation, prompt engineering, LLM orchestration (Ollama), embeddings, vector search, CLIP, DistilBERT, contrastive learning

Automation & APIs:
FastAPI, REST API design, Python scripting, Docker, CI/CD, workflow automation; ramping on no-code/low-code platforms (Make.com-style)

Full-Stack:
React, Vite, Tailwind CSS, React Query, PostgreSQL

Deep Learning & ML:
PyTorch, TensorFlow, Scikit-Learn, CNNs, EfficientNet, fine-tuning, timm, Hugging Face, NumPy, Pandas

Languages:
Python, SQL, PL/SQL, Core Java, C++

Tools:
Git/GitHub, pytest, Jupyter, IBM Watson

EDUCATION

MSc Data Science
University of Europe, Germany
Sep 2025 – Present

BTech – CS & Engineering
MLRIT, India
Jun 2018 – Jul 2022

CERTIFICATIONS

PL/SQL – Database Management

Core Java – Software Development

OneShield – P&C Insurance

LANGUAGES

English – Fluent

German – A1/A2 (learning)

PROFILE

AI Automation Engineer and Data Science MSc candidate who builds working AI solutions end-to-end — from stakeholder conversation to shipped tool. Built a framework-free RAG pipeline (no LangChain/LlamaIndex) exposing every retrieval mechanic, and a full-stack health app that turned operational tracking needs into a live logging, recommendation, and analytics system — backed by 2.5 years explaining technical tradeoffs to non-technical insurance stakeholders. Seeking a Working Student role on Seven Senders' AI Taskforce to turn cross-department friction into measurable, production AI solutions.

PROJECTS

RAG Pipeline – Retrieval-Augmented Generation from Scratch

Python · sentence-transformers · NumPy · Ollama · github.com/Deepak589/RAG_pipeline

- Built a framework-free RAG pipeline in ~270 lines of Python (no LangChain, LlamaIndex, or vector DB), implementing chunking, embedding, similarity search, and prompt assembly to keep every retrieval mechanic explicit and debuggable for handoff.
- Improved retrieval relevance by pairing a dense bi-encoder retriever (all-MiniLM-L6-v2, cosine top-3) with a hand-rolled TF-IDF baseline, plus heading-prefix chunking to cut off-topic matches.
- Reduced repeat embedding cost with a SHA-256-fingerprinted vector cache, and kept generation crash-free by degrading to retrieved-context output when the local LLM was offline — built for reliability, not just a demo.

RA Nutrition & Lifestyle App – Full-Stack Health Automation

React · FastAPI · PostgreSQL · Recharts · personal product build

- Translated operational health-tracking needs into a shipped full-stack feature: React/Vite/Tailwind frontend over a FastAPI + PostgreSQL backend (4 logical schemas), with JWT auth and token refresh.
- Built a rule-based recommendation engine and a weekly analytics dashboard (Recharts) that closes the loop from raw meal/symptom logs to actionable output for the end user.
- Architected (not yet shipped) an ML forecasting tier — scikit-learn/XGBoost and PyTorch LSTM/GRU/TCN — for flare-risk prediction, scoping the next iteration before building it.

Food Image-to-Recipe Retrieval – Multimodal Deep Learning

PyTorch · CLIP ViT-B/32 · DistilBERT · InfoNCE · MSc final project

- Built a dual-encoder retrieval model mapping food images (frozen CLIP ViT-B/32) and recipe text (DistilBERT) into a shared embedding space, reaching a validation median rank of 10/200 on 13.5K images.
- Shipped a reproducible, config-driven pipeline (OmegaConf/YAML, TensorBoard) with 58 unit tests and recall@k / median-rank evaluation — measuring what shipped, not just training it.

EXPERIENCE

Software Engineer Mar 2022 – Sep 2024
ValueMomentum

- Maintained 2 enterprise P&C insurance and surety-bond applications on OneShield PAS, resolving production defects across policy workflows.
- Built and optimized PL/SQL stored procedures and functions powering backend insurance data operations and endorsement modules.
- Translated business requirements from underwriting and claims stakeholders into technical specs during sprint planning, then accelerated releases by running QA/UAT test cases and supporting CI/CD deployment across the full policy lifecycle.

ML Intern Jun – Jul 2019
SmartBridge × IBM

- Built a lung-cancer risk-prediction ensemble (Random Forest, KNN, Gradient Boosting, Neural Network) and deployed it via IBM Watson ML, communicating results to non-technical reviewers with Matplotlib/Seaborn visualizations.